

1.0 PROBLEM STATEMENT

In Malaysia, there are many villages in rural areas that require power supply but without electrical connection from the utility grid. Perkampungan Orang Asli Kainpung Sungei Lui, Jempol Negeri Sembilan is one of them. The power supply is essential for the resident's daily needs and also to help their economies such as handicraft and herbs production. Besides that, the community of this village increase year by year hence the power generation required is increasing too. These reasons have placed the residents of 20 houses use standalone diesel generators as the choice of power supply since the village is far from the utility grid. Due to the continual increase of global fuel prices, people try to seek alternative power generation such as renewable energy

1.1 OBJECTIVE

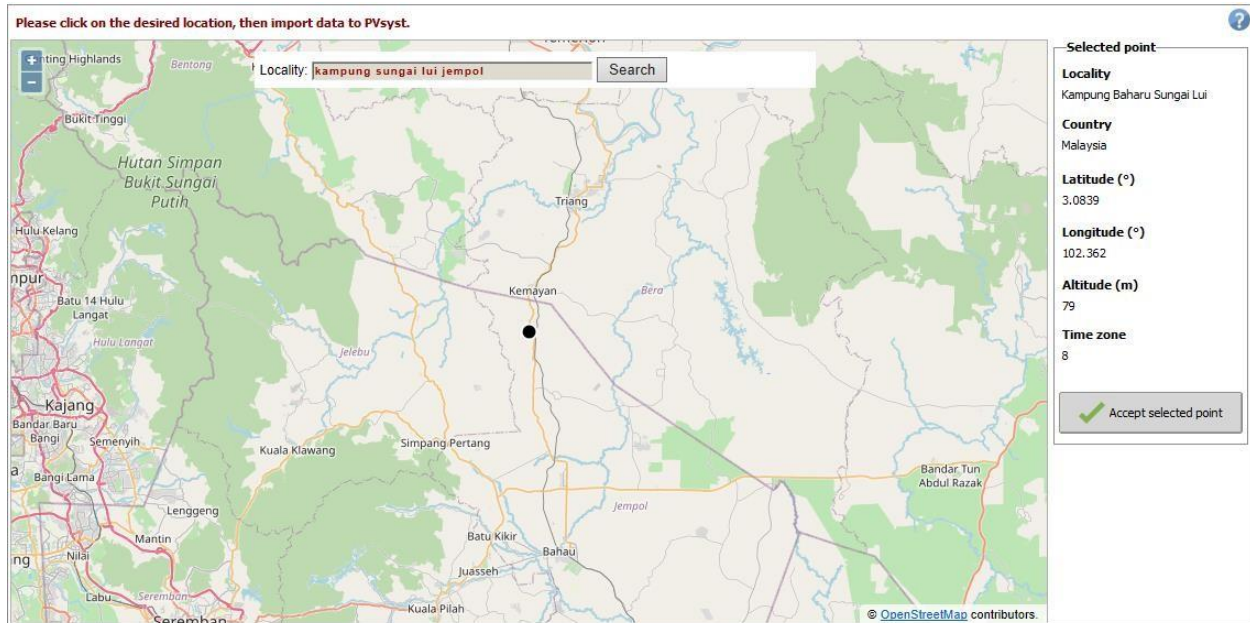
- The main point is to create perfect solar system for the local people
- A system that can help supply energy to the 20 houses by demand
- Invent an alternative way to generate electricity using renewable energy sources

1.2 SCOPE

- Use PVsyst and homer to design 20 houses.
- Project scope of this project is 20 houses in Perkampungan Orang Asli Kampung Sungei Lui, Jempol Negeri Sembilan.
- The typical load requirement for each other is 5 units of 40W lamps and 1 unit of 50W for the stand fan.
- Use Polycrystalline and Monocrystalline solar panel technology also use Lithium-ion and Lead acid battery technology.

2.0 METHOD TO SOLVING THE PROBLEM

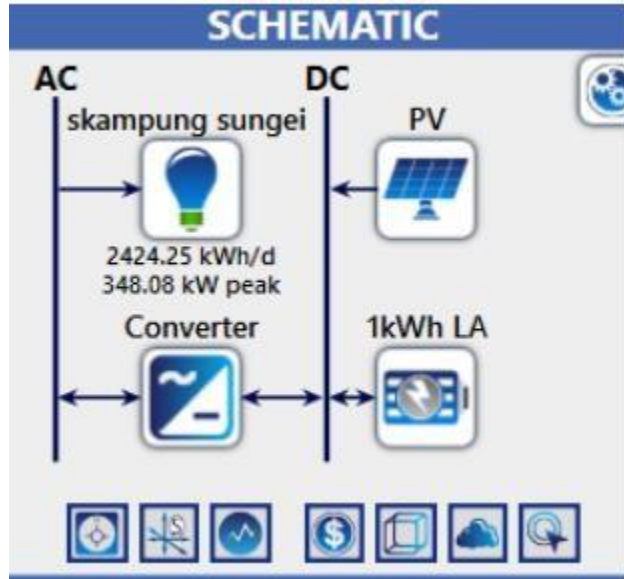
The remote location of the indigenous Sungai Lui village, or maybe this village itself, accounts for the lack of electricity in this community without the use of a diesel generator. In this village, electricity is produced by diesel generators, but as the cost of diesel rises, it is necessary to discover alternate sources of energy.



2.1 DESIGN THE SYSTEM BASED ON THE TYPICAL LOAD REQUIREMENT

To address this issue, we developed a method that allows local communities to generate electricity for daily consumption. To make things easier for them, we aim to build a solar farm to match the requirement for supplying electricity to 20 dwellings using 5 lamp units with 40W for six hours and 1 standing fan unit for 12 hours. This will not only help the community

3.0 CONSTRUCT ELECTRICAL CONNECTION



According to the Integration of Green Energy Resources for communities that are far from the grid connection the use of DC link is very appropriate. In addition, because the indigenous villages also take resources from DC, we choose to take to implement this DC link. In addition, this DC link is suitable for connecting between houses as well. There is also the benefit of DC link is that it does not require synchronization. Indirectly, it will make it easier for the Dc 20 home source system to work well. The DC link also has its advantages, although the terminal needs are more sophisticated, it also has a transmission infrastructure per kilometer that is cheaper and easier than AC. Indirectly, it does not require high costs and can transmit electricity easily. Next, if there is a residence that needs to use an ac load, you only need to add a dc to ac converter. Finally, because of the parallel source, high reliability if using a dc link

3.1 CALCULATION POLYCRYSTALLINE WITH ION

Poly - Lithium Ion	
① Lamp = $5 \times 40W = 200W$ Stand fan = $1 \times 50W = 50W$	② No of Battery $= \frac{1800}{265} = 6.7 \approx 7 \text{ battery}$
Total load = $250W$	
20 houses $250 \times 20 = 5000W (5kW)$	③ Charger controller size $= \frac{54 (250)}{30}$ $= 180 \text{ Amp}$
Lamp = $200W \times 6h = 1200Wh (1.2kWh)$ Stand fan = $50W \times 12h = 600Wh$	④ Inverter Size $1.25 \times 5k$ $= 6.25 kW$
Total energy = $1800Wh (1.8kWh)$	
$1800 \times 20 \text{ house} = 36000Wh (36kWh)$	
⑤ Consider loss = $1.5 \times 36000 = 54000Wh$	
⑥ Power output of PV panel = $\frac{54000Wh}{4h}$ $= 13500Wp$	
⑦ Quantity of PV panel $= \frac{13.5kWp}{250Wp} = 54 \text{ panel}$	
⑧ Size of battery $= \frac{54kWh}{30v} = 1800Ah$	

3.2 CALCULATION POLYCRYSTALLINE WITH LEAD ACID

Poly-lead acid

① Size of battery

$$\frac{54 \text{ kWh}}{25} = 2160 \text{ Ah}$$

② No of Battery

$$\frac{2160 \text{ Ah}}{250 \text{ Ah}} = 8.64 \approx 9 \text{ Battery}$$

③ Charger controller size

$$\frac{54 (250)}{25} = 540 \text{ Amp}$$

3.3 CALCULATION MONOCRYSTALLINE WITH LEAD ACID

mono-lead acid

① size of Battery

$$= \frac{54000}{27} \times 2000$$

② No of Battery

$$= \frac{2000}{255} \times 7.84 \approx 2 \text{ Battery}$$

③ Charger controller size

$$= \frac{54(250)}{27} = 500 \text{ Amp}$$

3.4 CALCULATION MONOCRYSTALLINE WITH LITHIUM ION

Mono lithium ion

① Size of Battery

$$= \frac{54000}{27} = 2000 \text{ Ah}$$

② No of Battery

$$= \frac{2000}{270} = 7.4 \approx 8 \text{ Battery}$$

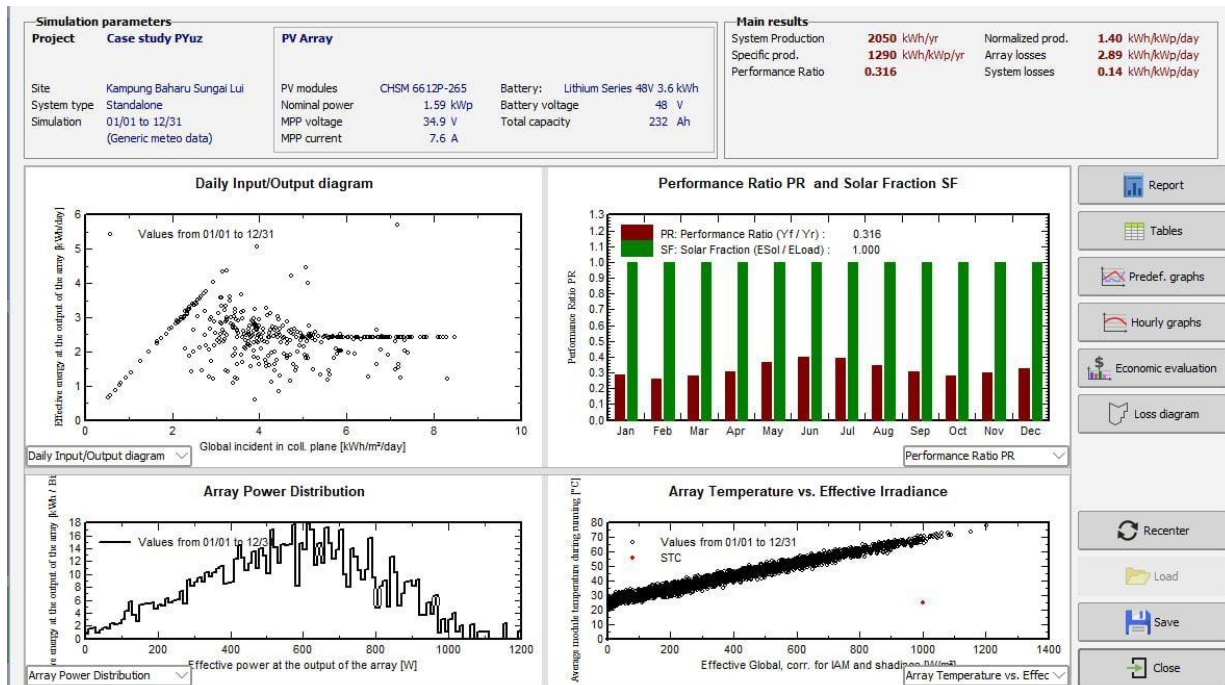
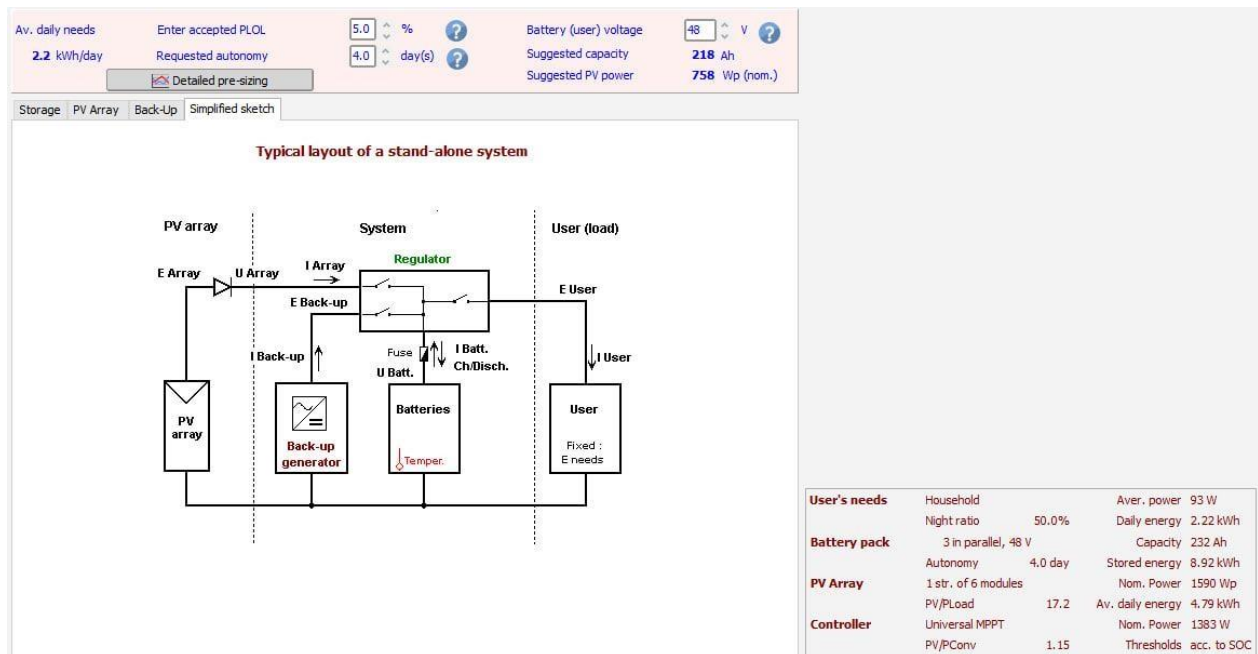
③ Charger controller size

$$= \frac{54 (250)}{27} = 500 \text{ Amp}$$

4.0 ANALYZE THE BEST PERFORMING

Before making a decision, we want to be sure that this community receives the greatest and most efficient solar system possible. Then we will conduct an analysis of the system that will be implemented.

Polycrystalline with lithium-ion



Balances and main results

	GlobHor kWh/m ²	GlobEff kWh/m ²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	134.2	147.9	193.7	119.9	0.000	68.94	68.94	1.000
February	143.1	147.4	190.9	126.4	0.000	62.27	62.27	1.000
March	162.4	149.3	195.2	123.3	0.000	68.94	68.94	1.000
April	160.8	130.5	171.5	102.8	0.000	66.72	66.72	1.000
May	156.6	112.8	149.3	76.4	0.000	68.94	68.94	1.000
June	147.3	99.6	133.6	64.6	0.000	66.72	66.72	1.000
July	151.3	105.4	141.6	69.5	0.000	68.94	68.94	1.000
August	154.4	119.1	158.7	87.6	0.000	68.94	68.94	1.000
September	150.6	133.1	175.5	106.1	0.000	66.72	66.72	1.000
October	149.7	149.0	194.3	122.5	0.000	68.94	68.94	1.000
November	127.8	137.7	180.2	110.6	0.000	66.72	66.72	1.000
December	118.1	128.7	166.0	94.0	0.000	68.94	68.94	1.000
Year	1756.3	1560.6	2050.5	1203.7	0.000	811.76	811.76	1.000

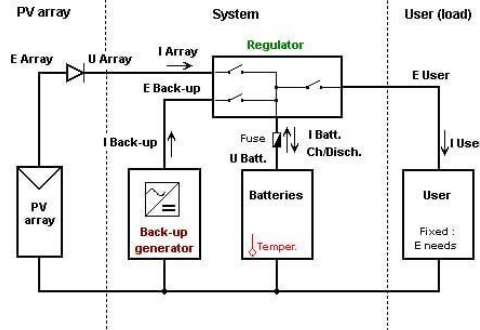
Polycrystalline with lead acid battery

Av. daily needs kWh/day
 Enter accepted PLOL %
 Requested autonomy day(s)
 Detailed pre-sizing

Battery (user) voltage V
 Suggested capacity **218** Ah
 Suggested PV power **750** Wp (nom.)

Storage PV Array Back-Up Simplified sketch

Typical layout of a stand-alone system



User's needs	Household	Aver. power	93 W
	Night ratio	Daily energy	2.22 kWh
Battery pack	3 in parallel, 48 V	Capacity	232 Ah
	Autonomy	Stored energy	8.92 kWh
PV Array	1 str. of 6 modules	Nom. Power	1500 Wp
	PV/Load	Av. daily energy	4.60 kWh
Controller	Universal MPPT	Nom. Power	1304 W
	PV/PConv	Thresholds	acc. to SOC

Simulation parameters

Project Case study PYuz

Site Kampung Baharu Sungai Lui
 System type Standalone
 Simulation 01/01 to 12/31
 (Generic meteo data)

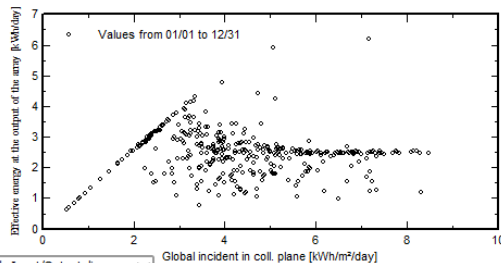
PV Array

PV modules Vision 250P60 frameless
 Nominal power 1.50 kWp
 MPP voltage 30.0 V
 MPP current 8.3 A
 Battery: S12-95 AGM
 Battery voltage 48 V
 Total capacity 230 Ah

Main results

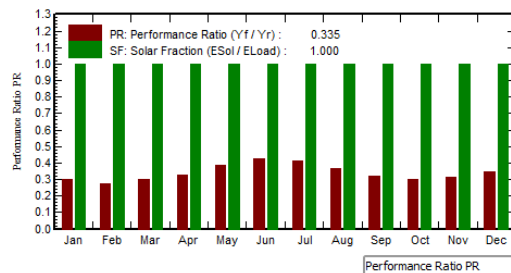
System Production **1941** kWh/yr
 Specific prod. **1294** kWh/kWp/yr
 Performance Ratio **0.335**
 Normalized prod. **1.48** kWh/kWp/day
 Array losses **2.76** kWh/kWp/day
 System losses **0.18** kWh/kWp/day

Daily Input/Output diagram



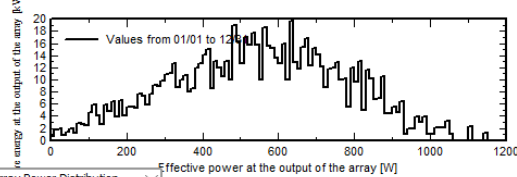
Daily Input/Output diagram Global incident in coll. plane [kWh/m²/day]

Performance Ratio PR and Solar Fraction SF



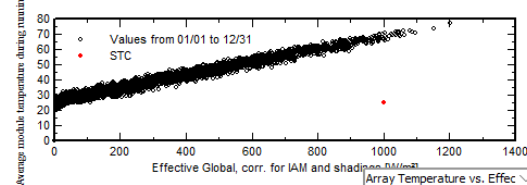
Performance Ratio PR

Array Power Distribution



Array Power Distribution Effective power at the output of the array [W]

Array Temperature vs. Effective Irradiance



Array Temperature vs. Effective Irradiance

Report

Tables

Predef. graphs

Hourly graphs

Economic evaluation

Loss diagram

Recenter

Load

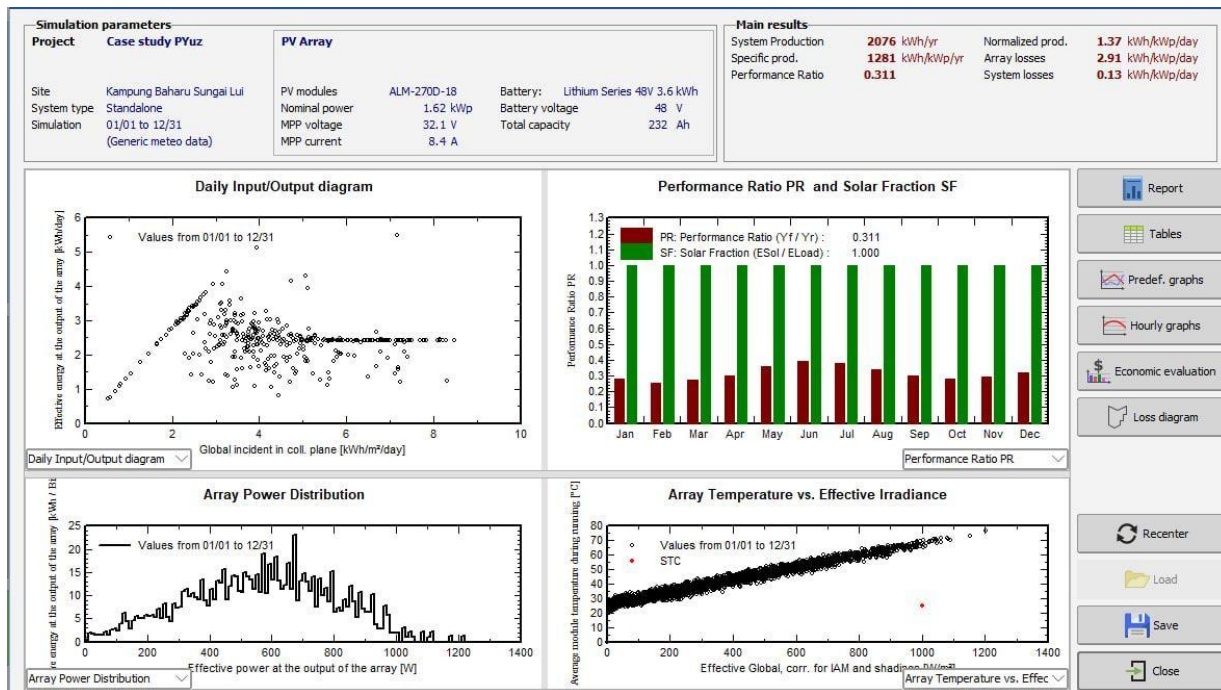
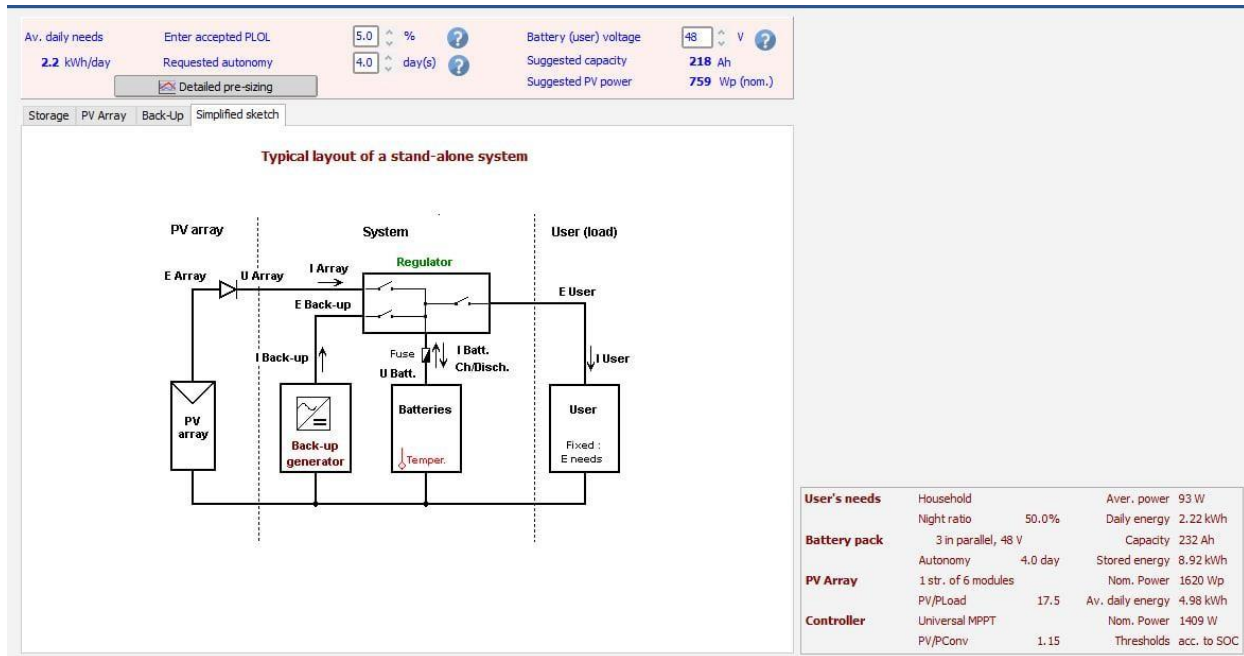
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Balances and main results

	GlobHor kWh/m ²	GlobEff kWh/m ²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	134.2	148.5	188.1	114.4	0.000	68.94	68.94	1.000
February	143.1	148.0	185.7	121.2	0.000	62.27	62.27	1.000
March	162.4	150.0	189.6	117.8	0.000	68.94	68.94	1.000
April	160.8	131.1	166.4	97.9	0.000	66.72	66.72	1.000
May	156.6	112.9	144.0	71.0	0.000	68.94	68.94	1.000
June	147.3	99.5	128.5	59.6	0.000	66.72	66.72	1.000
July	151.3	105.4	136.4	64.3	0.000	68.94	68.94	1.000
August	154.4	119.6	153.6	82.5	0.000	68.94	68.94	1.000
September	150.6	133.7	170.2	100.8	0.000	66.72	66.72	1.000
October	149.7	149.6	188.7	117.0	0.000	68.94	68.94	1.000
November	127.8	138.2	175.0	105.3	0.000	66.72	66.72	1.000
December	118.1	129.2	161.3	88.8	0.000	68.94	68.94	1.000
Year	1756.3	1565.9	1987.5	1140.6	0.000	811.76	811.76	1.000

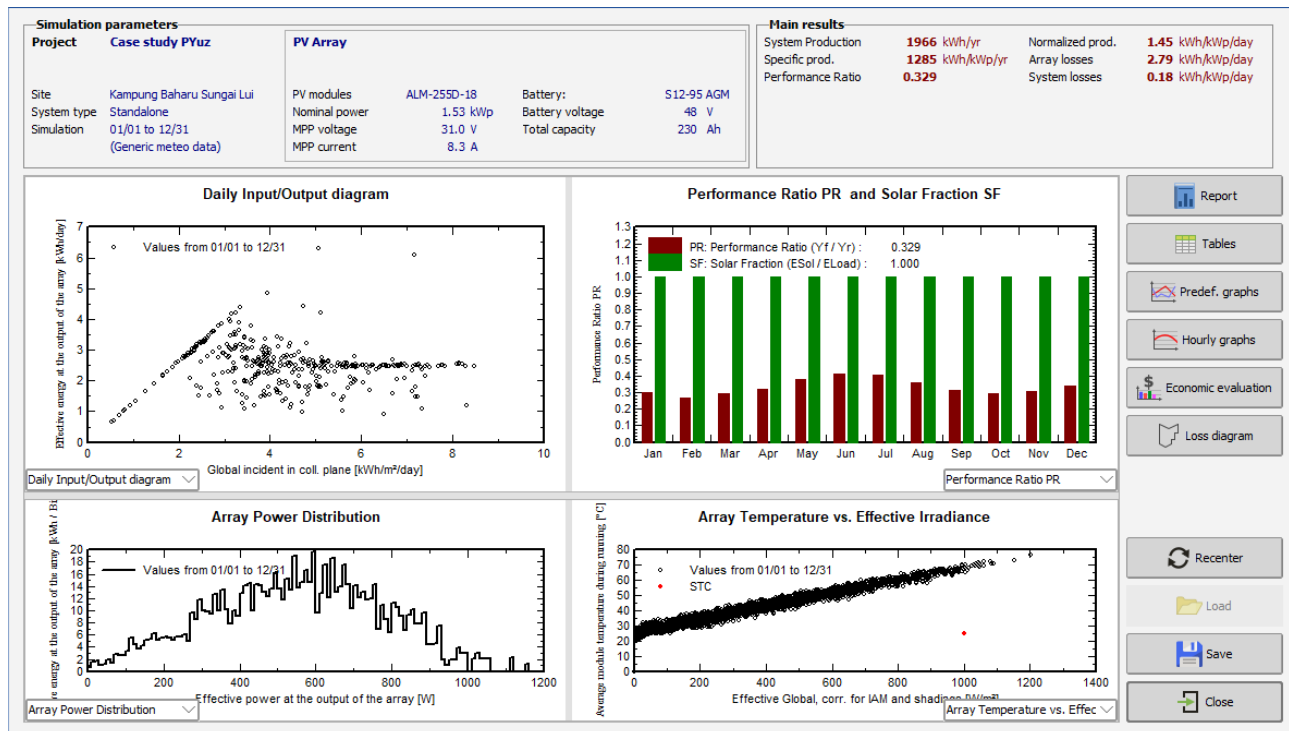
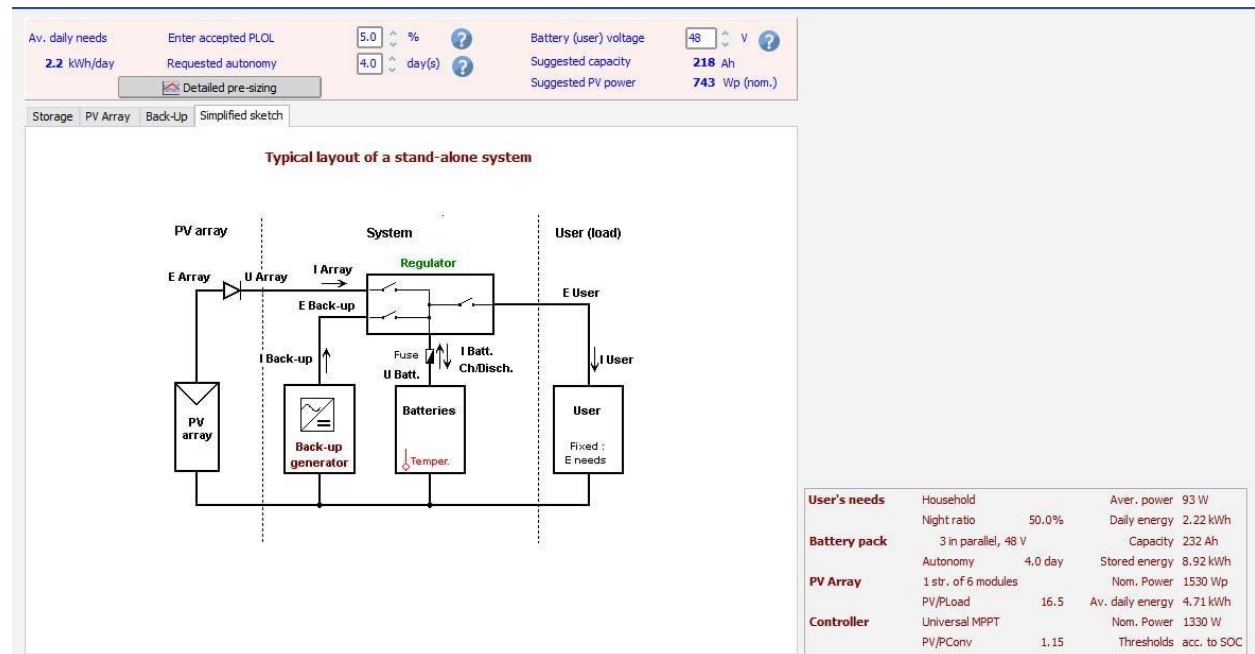
Monocrystalline with lithium-ion



Balances and main results

	GlobHor kWh/m ²	GlobEff kWh/m ²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	134.2	147.9	196.2	122.5	0.000	68.94	68.94	1.000
February	143.1	147.4	193.5	129.1	0.000	62.27	62.27	1.000
March	162.4	149.3	197.8	125.7	0.000	68.94	68.94	1.000
April	160.8	130.5	173.5	104.7	0.000	68.72	68.72	1.000
May	156.6	112.8	151.0	78.3	0.000	68.94	68.94	1.000
June	147.3	99.6	135.0	66.0	0.000	68.72	68.72	1.000
July	151.3	105.4	143.2	71.0	0.000	68.94	68.94	1.000
August	154.4	119.1	160.5	89.3	0.000	68.94	68.94	1.000
September	150.6	133.1	177.8	108.2	0.000	68.72	68.72	1.000
October	149.7	149.0	196.8	125.0	0.000	68.94	68.94	1.000
November	127.8	137.7	182.5	112.9	0.000	68.72	68.72	1.000
December	118.1	128.7	168.6	96.5	0.000	68.94	68.94	1.000
Year	1756.3	1560.6	2075.9	1229.3	0.000	811.76	811.76	1.000

Monocrystalline with lead acid battery



Balances and main results

	GlobHor kWh/m ²	GlobEff kWh/m ²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	134.2	147.9	185.9	112.2	0.000	68.94	68.94	1.000
February	143.1	147.4	183.3	118.8	0.000	62.27	62.27	1.000
March	162.4	149.3	187.2	115.4	0.000	68.94	68.94	1.000
April	160.8	130.5	164.4	95.9	0.000	66.72	66.72	1.000
May	156.6	112.8	143.0	70.0	0.000	68.94	68.94	1.000
June	147.3	99.6	128.0	59.1	0.000	66.72	66.72	1.000
July	151.3	105.4	135.6	62.9	0.000	68.94	68.94	1.000
August	154.4	119.1	152.0	81.6	0.000	68.94	68.94	1.000
September	150.6	133.1	168.2	98.9	0.000	66.72	66.72	1.000
October	149.7	149.0	186.4	114.6	0.000	68.94	68.94	1.000
November	127.8	137.7	172.9	103.2	0.000	66.72	66.72	1.000
December	118.1	128.7	159.3	86.3	0.000	68.94	68.94	1.000
Year	1756.3	1560.6	1966.3	1118.9	0.000	811.76	811.76	1.000

THE BEST PERFORMING

BATTERY OF LITHIUM ION WITH POLYCRYSTALLINE

PV Array Characteristics			
PV module		Battery	
Manufacturer	Astronergy	Manufacturer	Turbo Energy
Model	CHSM 6612P-265	Model	Lithium Series 48V 3.6 kWh
(Original PVsyst database)		Technology	Lithium-ion, LFP
Unit Nom. Power	265 Wp	Nb. of units	3 in parallel
Number of PV modules	6 units	Discharging min. SOC	20.0 %
Nominal (STC)	1590 Wp	Stored energy	8.7 kWh
Modules	1 String x 6 In series	Battery Pack Characteristics	
At operating cond. (50°C)		Voltage	48 V
Pmpp	1419 Wp	Nominal Capacity	232 Ah (C10)
U mpp	183 V	Temperature	Fixed 20 °C
I mpp	7.7 A	Battery Management control	
Controller		Threshold commands as	SOC calculation
Universal controller		Charging	SOC = 0.92 / 0.75
Technology	MPPT converter	Discharging	SOC = 0.20 / 0.45
Temp coeff.	-5.0 mV/°C/Elem.		
Converter			
Maxi and EURO efficiencies	97.0 / 95.0 %		
Total PV power			
Nominal (STC)	1.59 kWp		
Total	6 modules		
Module area	11.7 m²		
Cell area	10.5 m²		

BATTERY OF LEAD ACID WITH POLYCRYSTALLINE

PV Array Characteristics			
PV module		Battery	
Manufacturer	Yingli Solar	Manufacturer	Turbo Energy
Model	YL250P-29b	Model	Lithium Series 48V 3.6 kWh
(Original PVsyst database)		Technology	Lithium-ion, LFP
Unit Nom. Power	250 Wp	Nb. of units	3 in parallel
Number of PV modules	6 units	Discharging min. SOC	20.0 %
Nominal (STC)	1500 Wp	Stored energy	8.7 kWh
Modules	1 String x 6 In series	Battery Pack Characteristics	
At operating cond. (50°C)		Voltage	48 V
Pmpp	1340 Wp	Nominal Capacity	232 Ah (C10)
U mpp	162 V	Temperature	Fixed 20 °C
I mpp	8.3 A	Battery Management control	
Controller		Threshold commands as	SOC calculation
Universal controller		Charging	SOC = 0.92 / 0.75
Technology	MPPT converter	Discharging	SOC = 0.20 / 0.45
Temp coeff.	-5.0 mV/°C/Elem.		
Converter			
Maxi and EURO efficiencies	97.0 / 95.0 %		
Total PV power			
Nominal (STC)	1.50 kWp		
Total	6 modules		
Module area	9.7 m²		
Cell area	8.8 m²		

BATTERY OF LITHIUM-ION WITH MONOCRYSTALLINE

PV Array Characteristics			
PV module		Battery	
Manufacturer	Alex Solar	Manufacturer	Turbo Energy
Model	ALM-270D-18	Model	Lithium Series 48V 3.6 kWh
(Original PVsyst database)		Technology	Lithium-ion, LFP
Unit Nom. Power	270 Wp	Nb. of units	3 in parallel
Number of PV modules	6 units	Discharging min. SOC	20.0 %
Nominal (STC)	1620 Wp	Stored energy	8.7 kWh
Modules	1 String x 6 In series	Battery Pack Characteristics	
At operating cond. (50°C)		Voltage	48 V
P _{mpp}	1439 Wp	Nominal Capacity	232 Ah (C10)
U _{mpp}	170 V	Temperature	Fixed 20 °C
I _{mpp}	8.5 A		
Controller		Battery Management control	
Universal controller		Threshold commands as	SOC calculation
Technology	MPPT converter	Charging	SOC = 0.92 / 0.75
Temp coeff.	-5.0 mV/°C/Elem.	Discharging	SOC = 0.20 / 0.45
Converter			
Maxi and EURO efficiencies	97.0 / 95.0 %		
Total PV power			
Nominal (STC)	1.62 kWp		
Total	6 modules		
Module area	9.7 m²		
Cell area	8.6 m²		

BATTERY OF LEAD ACID WITH MONOCRYSTALLINE

PV Array Characteristics			
PV module		Battery	
Manufacturer	Alex Solar	Manufacturer	Turbo Energy
Model	ALM-255D-18	Model	Lithium Series 48V 3.6 kWh
(Original PVsyst database)		Technology	Lithium-ion, LFP
Unit Nom. Power	255 Wp	Nb. of units	3 in parallel
Number of PV modules	6 units	Discharging min. SOC	20.0 %
Nominal (STC)	1530 Wp	Stored energy	8.7 kWh
Modules	1 String x 6 In series	Battery Pack Characteristics	
At operating cond. (50°C)		Voltage	48 V
P _{mpp}	1362 Wp	Nominal Capacity	232 Ah (C10)
U _{mpp}	164 V	Temperature	Fixed 20 °C
I _{mpp}	8.3 A		
Controller		Battery Management control	
Universal controller		Threshold commands as	SOC calculation
Technology	MPPT converter	Charging	SOC = 0.92 / 0.75
Temp coeff.	-5.0 mV/°C/Elem.	Discharging	SOC = 0.20 / 0.45
Converter			
Maxi and EURO efficiencies	97.0 / 95.0 %		
Total PV power			
Nominal (STC)	1.53 kWp		
Total	6 modules		
Module area	9.7 m²		
Cell area	8.6 m²		

4.1 COMPARISON OF POLYCRYSTALLINE AND MONOCRYSTALLINE

MONOCRYSTALLINE	POLYCRYSTALLINE
Pros ▪ Highly efficient at producing energy ▪ Panels require less space ▪ Black panels blend with darker shingles or foliage ▪ Better heat tolerance	Pros ▪ Less expensive than monocrystalline panels ▪ Lifespan comparable to that of monocrystalline panels yet at a lower cost
Cons ▪ Expensive ▪ Less sustainable production methods	Cons ▪ Panels require more space ▪ Less efficient at producing energy ▪ Less sustainable production ▪ The bluish color stands out more than the black of the polycrystalline panels ▪ Less heat tolerance

- According to our findings, monocrystalline is far better than polycrystalline since monocrystalline is more efficient in generating electricity than polycrystalline. To achieve the requirements, we require an efficient solar panel capable of generating solar energy at a high rate.
- Additionally, monocrystals are better at generating energy at a higher pace than polycrystals, which have relatively low heat tolerance. This is because they have superior heat tolerance. This is because the solar panel's efficiency may somewhat decline when its surface temperature reaches this level.

4.2 COMPARISON OF LEAD ACID AND LITHIUM

	Lithium-ion	Lead acid
Capacity	15+ kWh	1.5-5kWh
Depth of discharge	85%	50%
Efficiency	95%	80-85%
Lifespan	10-15 years	3-12 years

- If you have used it more than 50%, the battery's life cycle will get affected. The depth of discharge is 50% for lead-acid batteries, and 80% for lithium batteries, respectively. This means you can use the lithium-ion battery for a longer period without recharging
- Following that lithium ion battery have a longer lifespan compare to lead acid this will able the local to use it longer and don't need to change it frequently.

COMPARISON FOR THE SOLAR PANELS AND BATTERIES

Comparison	<i>Poly-lithium</i>	<i>Poly-lead acid</i>	<i>Mono-lithium</i>	<i>Mono-lead acid</i>
System production	2050 kWh/year	1941 kWh/year	2076 kWh/year	1966 kWh/year
Performance ratio	0.316	0.335	0.311	0.329
Available energy	2050 kWh/year	1941 kWh/year	2076 kWh/year	1966 kWh/year
Used energy	666.81 kWh/year	666.81 kWh/year	666.81 kWh/year	666.81 kWh/year
Solar fraction	100%	100%	100%	100%

CONCLUSION

In conclusion, from this case study we were able to reduce costs for the residents of the indigenous community in the Indigenous Village, Sungai Lui by using the DC link. Also, the benefit of a DC link is that it does not require synchronization. In addition, with the use of solar energy it is possible to avoid the occurrence of noise or air pollution and so on. With the use of the system, it is able to supply energy to 20 houses. Finally, with the use of solar energy sources as the main energy, it is able to provide comfort to them and also benefits in the future.